

WASTE → VALUE

CharLoop

Husk → biochar

Mill-scale rice-husk pyrolysis yields durable biochar credits, heat, and gas-independence



THE SITUATION IN BANGLADESH

Bangladesh's industrial backbone is starving for energy: roughly Tk35,000 crore of private investment sits idle across economic zones, and nearly 400 gas-dependent factories cannot run at full capacity for want of gas ⁵. Domestic fields are drying up — supply has fallen toward 2,500 mmcf against demand near 3.8 bcf — leaving the country dependent on costly LNG that industries pay Tk30 for while imports cost around Tk100 per unit ⁵. Yet the same country is the world's third-largest rice producer, milling about 39 million tonnes a year and generating roughly 7.5–9 million tonnes of rice husk, a residue that is 22% of paddy weight and concentrated at the mill gate ². Today most of that husk is simply burned for crude parboiling steam or dumped, wasting both its energy and its carbon value ⁴.

OUR PITCH

CharLoop installs small, portable pyrolysis units at rice mills that convert this concentrated husk into process heat first — replacing scarce, expensive gas with on-site energy independence the mill already feeds itself ⁴. The same reaction yields biochar, a soil amendment whose fused-aromatic carbon resists decomposition for centuries (IPCC-recognized 100–1,000-year permanence) and lifts crop yields by roughly 10–16% on average ¹⁶. That biochar is also a durable carbon-removal credit: biochar already accounts for over 90% of CDR tonnes delivered, certifiable under Puro.earth's audited CORC methodology at steady 2025 prices of about \$125–\$145 per tonne CO₂ ⁸⁹³. The model is proven — MIT spinoff Takachar won the 2021 Earthshot Prize deploying near-identical portable husk pyrolysis, cutting open-burn smoke up to 98% and raising farmer income 30–50% ⁷. CharLoop's honest wedge is energy first, carbon as upside: the heat savings justify the unit even before a single credit sells ⁴⁵.

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